

September 4, 2019

Mr. Doug Langworthy
City of Cadillac WWTP
200 N. Lake Street
Cadillac, Michigan 49601



Reference: 0501867.0106

Subject: Whole Effluent Toxicity Test Results

Dear Doug,

Enclosed please find the final results of the following Chronic Toxicity Test performed on samples of the City of Cadillac WWTP Outfall 001A effluent.

- 13 August 2019, Chronic *Ceriodaphnia dubia* Toxicity Test

If you have any questions concerning this report or if I can be of any further assistance to you, please feel free to contact me at (616) 738-7308 or via e-mail at bruce.rabe@erm.com.

Yours sincerely,

Bruce A. Rabe
Director, Aquatic Toxicology Laboratory

BAR:km

Enclosures: Whole Effluent Toxicity Test Report

cc: Cindy Tomaszewski
File

FINAL REPORT

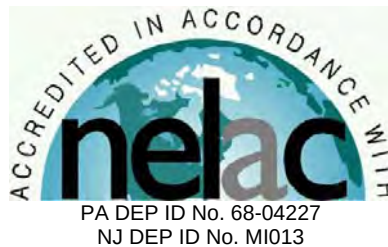
Chronic Toxicity Test
Freshwater Invertebrate,
Ceriodaphnia dubia
EPA Test Method 1002.0

Submitted To:
City of Cadillac WWTP
200 N. Lake Street
Cadillac, MI 49601

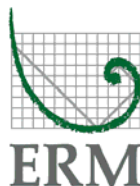
Sample: Outfall 001A

Testing Period: 13 – 19 August 2019

Laboratory I.D. Number: 081319-10



Conducted By:
Environmental Resources Management, Inc.
3352 128th Avenue
Holland, Michigan 49424



Test Overview



Permittee: City of Cadillac WWTP
Location: 1121 Plett Rd.
Cadillac, MI 49601
Contact: Mr. Doug Langworthy
Telephone #: 231.878.4437

NPDES Permit #: MI0020257
Permit Requirements: Acute Toxicity = Report Only
Chronic Toxicity Limit = 1.1 TUc

Test Sample: Outfall 001A
Receiving Water: Clam River

Testing Date: 13 – 19 August 2019

Sample Date(s): 12 August 2019
14 August 2019
16 August 2019

Test/Method: Daphnid, *Ceriodaphnia dubia*,
Survival and Reproduction Test
EPA 821-R-02-013 Method
1002.0.

QC Objectives: Test data met all test
acceptability criteria, except
where noted below.

Data Qualifiers: None

DATA SUMMARY

Effluent Concentrations (%)	Survival (%)	Reproduction (Average Young/Female)
Control	100	25.6
13	100	36.6
25	100	30.4
50	100	19.5
76	100	14.8*
100	100	12.8*

* Significantly lower than the control (P=0.01)

TEST RESULTS

48-Hour LC ₅₀	>100%
NOEC	50%
LOEC	76%
ChV	61.6%
MSDp	30.5%
Acute Toxicity Unit (TUa)	0
Chronic Toxicity Unit (TUc)	1.6

TEST CONCLUSION

In accordance with the NPDES permit requirement for the City of Cadillac WWTP, this toxicity test did not exhibit acute toxicity but did exceed the chronic toxicity limit.

Bruce A. Rabe
Director, Aquatic
Toxicology Laboratory
ERM Project No. 0501867.0106

Environmental Resources Management
3352 128th Avenue
Holland, Michigan 49424-9263
Phone: 616.399.3500
Fax: 616.399.3777



ERM Testing Method

Ceriodaphnia dubia – Survival and Reproduction Toxicity Test



Upon sample receipt, each effluent sample was analyzed for a suite of water quality parameters (Appendix A - Table 1). Where indigenous organisms were present, the sample was filtered through a 60 micron (µm) NITEX® screen. All samples were maintained at 0 – 6 degrees Celsius (°C) until needed for testing.

A series of five effluent concentrations and a control solution were established for testing. All test solutions were prepared by mixing appropriate volumes of dilution water and effluent in the test containers. Dilution water consisted of reconstituted moderately hard water. The control solution consisted of 100 percent dilution water.

Ceriodaphnia dubia used to initiate this test were obtained from individual, in-house cultures and were less than 24-hours old, and had an age range of 0 to 8 hours at test initiation. Test organisms used to initiate this test were released from adults which met acceptable performance criteria (i.e., ≥15 young/surviving female within 3 broods and obtained from a brood of at least 8 young) and were maintained in reconstituted moderately hard water prior to test initiation.

The *Ceriodaphnia dubia* test was conducted using 30-milliliter (mL) disposable polystyrene containers containing 15 mL of control water or test solution. One *Ceriodaphnia dubia* was added to each test chamber with ten replicate chambers per treatment. Each *Ceriodaphnia dubia* test chamber was fed a 0.2-mL suspension consisting of yeast-Cerophyll-trout chow (YCT) and green algae (*Raphidocelis subcapitata*) mixture daily.

The test solutions were renewed daily during the exposure by transferring the adult daphnid, by way of a wide bore pipette, into fresh control water or test solution.

Percent survival of exposed *Ceriodaphnia dubia* was determined by inspecting for adult mortality daily. Mortality was defined as no body or appendage movement after gentle prodding. Production of young was also determined by daily inspections and enumeration. When 60 percent of the surviving females in the control treatment produced three broods, mean reproduction was determined by calculating the average number of live young produced per female for each treatment.

The test was conducted at a temperature of 25 ± 1°C under fluorescent lighting with a photoperiod of 16 hours light and 8 hours dark. Water quality measurements were performed on all control and test solutions prior to test initiation and on selected treatments daily thereafter, as indicated in the raw data (Appendix A - Table 2).

Following termination of the chronic toxicity test, No Observed Effect Concentrations (NOEC) and Lowest Observed Effect Concentrations (LOEC) were determined for *Ceriodaphnia dubia* survival and reproduction. An NOEC is defined as the highest effluent concentration that does not produce any observed adverse effect to the exposed test organism. An LOEC is defined as the lowest effluent concentration that does produce an observed adverse effect to the exposed test organism. An adverse effect is determined as a statistically significant difference between the control and a given effluent concentration. Significant differences in *Ceriodaphnia dubia* survival were determined using the Fisher's Exact Test.

Prior to the determination of any significant differences in *Ceriodaphnia dubia* reproduction, the data were evaluated for normal distribution and homogeneity characteristics. Depending on the result and the number of test replicates per concentration, an analysis of variance test was performed followed by one of the following mean comparison tests: Dunnett's Procedure, Bonferroni t-Test, Steel's Many-One Rank Test, Wilcoxon Rank Sum Test, or the T-Test. Based on the NOEC and LOEC determinations, a chronic value (ChV) was calculated for the most sensitive of the two test endpoints (i.e., survival or reproduction) and converted to a chronic toxic unit (TUc) for reporting purposes by $100/\text{ChV}$, unless the LOEC was greater than 100 percent in which case the TUc was reported as 0.

To evaluate acute toxicity, a 48-hour LC_{50} and corresponding 95 percent confidence interval was also calculated, where possible. The LC_{50} value estimate was determined by using one of the following statistical methods: graphical, Spearman-Kärber, Trimmed Spearman-Kärber, or Probit. The method selected for reporting test results was determined by the characteristics of the data; that is, the presence or absence of 0 and 100 percent mortality and the number of concentrations in which mortalities between 0 and 100 percent occurred. For reporting purposes, the 48-hour LC_{50} value was converted to an acute toxic unit (TUa) by $100/\text{LC}_{50}$.

For 48-hour LC_{50} values greater than 100 percent, the TUa was derived as follows: (1) if mortality in the 100 percent effluent was 0 percent to 10 percent, the TUa was reported as 0, (2) if mortality in the 100 percent effluent was 11 percent to 49 percent, the TUa was calculated and reported as 0.02 times the percent mortality in 100 percent effluent. All statistical analyses were performed using the CETIS™ Version 1.9.4.3 software program.

The reference toxicant, sodium chloride, was used to monitor the sensitivity of the test organisms and the precision of the testing procedure. Chronic reference toxicant tests are performed at least monthly and the resulting Inhibition Concentrations (IC_{25}) are plotted to determine if the results are within prescribed limits (Appendix A - Standard Reference Toxicant Data). If the IC_{25} of a particular reference toxicant test does not fall within the expected range of $\pm$ two standard deviations from the mean for a given test organism, the sensitivity of that organism and the overall credibility of the test system is suspect.

Reference:

USEPA. 2002. Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms, 4th Ed. U.S. Environmental Protection Agency, Office of Water, Washington, D.C., EPA-821-R-02-013.

Case Narrative



1.0 TEST PERFORMANCE CRITERIA

The quality control results achieved laboratory specifications.

2.0 MODIFICATIONS TO ERM'S STANDARD TEST METHOD

Test was performed in accordance with ERM's standard test method (see page 3).

Appendix A
Supporting Documents

- *Raw Test Data*
- *Statistical Analysis (if necessary)*
- *Chain-of-Custody Forms*
- *Standard Reference Toxicant Data*

**Environmental
Resources
Management**

***Ceriodaphnia dubia* - Chronic Toxicity Test
Initial Water Quality and Test Solution Preparation**

Table 1
Page 1 of 1

Permittee/Client:	City of Cadillac WWTP	Control/Dilution Water:	RMHW
Effluent/Location:	Outfall 001A	Organism Batch #:	140-19
Lab I.D.#:	081319-10	Organism Age:	15-20.5 hrs
Beginning Date:	08/13/19	QC Review:	KM
Ending Date:	08/19/19	QC Review Date:	08/21/19
	Time: 1300		
	Time: 1330		

Initial Water Quality:

Parameter	Units	Effluent			Synthetic Water		
Sample #	--	1	2	3	--	--	--
Lab I.D.#/ Batch #	--	081319-10	081519-3	081719-10	92-19	96-19	-
Temperature	° C	2	2	1	--	--	--
Dissolved Oxygen	mg / L	10.1	10.7	10.2	--	--	--
pH	S.U.	7.3	7.6	7.1	8.0	7.8	-
Conductivity	umhos/cm	995	974	1017	320	321	-
Alkalinity	mg / L CaCO ₃	76	72	78	60	60	-
Hardness	mg / L CaCO ₃	140	160	180	84	84	-
Total Ammonia	mg / L NH ₃	<0.01	<0.01	<0.01	--	--	--
Total Residual Chlorine	mg / L Cl ₂	<0.01	<0.01	<0.01	<0.01	<0.01	-
Total mls of Sodium Thiosulfate added per liter	mg / L	--	--	--	--	--	--
Initials	--	MS	MS	SMR	MS	MS	-

Treatment (% Effluent)	Effluent (mL)	Dilution (mL)	Test Day	Initials	Effluent Sample #	Synthetic Batch #
Control	0	200	0	PH	1	92-19
13%	26	174	1	PH	1	92-19
25%	50	150	2	MS	2	96-19
50%	100	100	3	PH	2	96-19
76%	152	48	4	SMR	3	96-19
100%	200	0	5	SMR	3	96-19
			6	PH	-	-
			7			

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 081319-10

Water Quality Data:

Dissolved Oxygen (mg/L)														
Day														
Meter #	5	5	5	5	5	5	5	3	3	3	3	5		
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	7.9	7.8	7.8	7.8	8.0	7.8	8.1	8.1	8.3	7.7	8.2	7.7		
13%	8.0	7.8	7.8	7.8	8.0	7.9	8.1	8.2	8.4	7.8	8.2	7.9		
25%	8.0	7.9	7.8	7.8	8.0	7.9	8.1	8.2	8.4	7.8	8.2	7.7		
50%	8.0	7.9	7.8	7.8	8.0	8.0	8.1	8.3	8.4	7.8	8.2	8.0		
76%	8.0	7.9	7.8	7.8	8.0	7.9	8.1	8.3	8.4	7.8	8.3	8.0		
100%	8.0	7.9	7.8	7.8	8.0	7.9	8.0	8.3	8.5	7.8	8.3	7.7		
pH (S.U.)														
Day														
Meter #	10	10	10	9	8	10	10	9	9	9	9	10		
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	7.9	7.9	7.9	7.8	7.8	7.7	7.9	7.5	7.9	7.3	8.0	8.0		
13%	--	8.0	--	8.0	--	7.9	--	7.7	--	7.4	--	8.0	--	
25%	--	8.0	--	8.0	--	8.0	--	7.9	--	7.6	--	8.0	--	
50%	--	8.1	--	8.0	--	8.1	--	8.1	--	7.8	--	8.1	--	
76%	--	8.2	--	8.1	--	8.2	--	8.1	--	7.9	--	8.1	--	
100%	7.3	8.2	7.7	8.2	7.7	8.2	7.8	8.2	7.7	8.1	7.6	8.2		
Conductivity (umhos / cm)														
Day														
Meter #	4	--	4	--	4	--	4	--	4	--	4	--		--
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	322	--	322	--	320	--	323	--	334	--	319	--		--
13%	400	--	401	--	403	--	401	--	414	--	405	--		--
25%	478	--	482	--	483	--	487	--	503	--	489	--		--
50%	637	--	640	--	641	--	646	--	675	--	656	--		--
76%	801	--	814	--	811	--	813	--	854	--	827	--		--
100%	957	--	968	--	969	--	956	--	1025	--	995	--		--
Temperature (° C)														
Day														
Meter #	5	5	5	5	5	5	5	3	3	3	3	5		
Treatment	0	1		2		3		4		5		6		7
(% Effluent)	I	F	I	F	I	F	I	F	I	F	I	F	I	F
Control	24	24	24	24	24	24	24	24	24	25	24	24		
13%	24	24	24	24	24	24	24	24	24	25	24	24		
25%	24	24	24	24	24	24	24	24	24	25	24	24		
50%	25	24	24	24	24	24	24	24	24	25	24	24		
76%	25	24	24	24	24	24	24	24	24	25	25	24		
100%	25	24	24	24	24	24	25	24	25	25	25	24		
I = Initial Chemistry F = Final Chemistry														

Note: D.O. meter also used for temperature measurement unless otherwise noted.

Environmental
Resources
Management

Ceriodaphnia dubia - Chronic Toxicity Test
Survival and Reproduction Data

Table 3
Page 1 of 2

Permittee/Client: City of Cadillac WWTP
Effluent/Location: Outfall 001A
Lab I.D.#: 081319-10

Treatment (% Effluent)	Day No.	Replicate										Average Young/ Female	Number of Live Adults (% Sur.)	Average Young/ Female % CV
		1	2	3	4	5	6	7	8	9	10			
Control	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	5	6	4	4	6	6	4	5	5	6		10	
	4	5	6	4	4	6	6	4	5	5	6		10	
	5	12	10	8	11	5	11	9	6	12	15		10	
	6	4	14	16	18	14	17	13	3	16	1		10	
	7													
Totals:		21	30	28	33	15	34	26	14	33	22	25.6	(100)	28.7
# Broods (% 3rd Brood)		3	3	3	3	3	3	3	3	3	3	(100)		
13%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	6	5	6	6	6	6	6	5	5	5		10	
	4	5	11	13	12	13	13	10	5	10	1		10	
	5	16	1	1	1	13	1	7	7	11	11		10	
	6	12	20	19	21	22	21	23	12	21	17		10	
	7	34	36	38	39	41	40	45	24	36	33	36.6 03/14/14	(100)	15.5
Totals:		34	36	38	39	41	40	45	24	36	33	36.6	(100)	15.5
25%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	5	5	5	5	6	6	6	6	7	7		10	
	4	6	5	5	9	4	4	4	4	7	7		10	
	5	10	10	7	10	7	7	10	10	12	13		10	
	6	8	15	14	18	18	14	14	11	18	20		10	
	7													
Totals:		24	30	31	32	32	27	30	22	36	40	30.4	(100)	17.4
50%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	5	5	5	5	6	6	6	6	7	7		10	
	4	6	5	5	9	4	4	4	4	7	7		10	
	5	10	10	7	10	7	7	10	10	12	13		10	
	6	8	15	14	18	18	14	14	11	18	20		10	
	7													
Totals:		15	24	23	20	24	19	19	7	18	26	19.5	(100)	28.3
76%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	5	5	5	5	6	6	6	6	7	7		10	
	4	2	3	3	2	3	3	4	2	1	2		10	
	5	4	5	6	6	4	4	5	5	4	4		10	
	6	5	11	12	10	11	9	7	5	7	9		10	
	7													
Totals:		16	19	21	18	18	16	16	7	12	15	14.8	(100)	33.8
100%	1	--	--	--	--	--	--	--	--	--	--		10	
	2	--	--	--	--	--	--	--	--	--	--		10	
	3	5	5	5	5	6	6	6	6	7	7		10	
	4	1	2	2	2	2	2	2	1	2	1		10	
	5	2	3	6	5	4	4	4	3	3	4		10	
	6	5	11	12	10	11	9	7	5	7	9		10	
	7													
Totals:		3	16	21	15	16	16	17	4	15	5	12.8	(100)	49.4

X = DEAD ADULT 1X = DEAD ADULT, ONE YOUNG PRODUCED BEFORE DEATH -- = NO YOUNG RECORDED
(E) = ABORTED EMBRYOS /EGGS (1) = ONE DEAD YOUNG (S) = SPLIT BROOD * = 4th BROOD EXCLUDED FROM TOTAL

CETIS Analytical Report

Report Date: 19 Aug-19 16:11 (p 1 of 2)
 Test Code/ID: 4236F9AA / 11-1089-9114

Ceriodaphnia 7-d Survival and Reproduction Test

ERM

Analysis ID: 13-3335-8959	Endpoint: Reproduction	CETIS Version: CETISv1.9.4
Analyzed: 19 Aug-19 16:09	Analysis: Parametric-Control vs Treatments	Status Level: 1
Batch ID: 07-9941-3578	Test Type: Reproduction-Survival (7d)	Analyst: Lab Tech
Start Date: 13 Aug-19 13:00	Protocol: EPA/821/R-02-013 (2002)	Diluent: Reconstituted Water
Ending Date: 19 Aug-19 13:30	Species: Ceriodaphnia dubia	Brine:
Test Length: 6d 1h	Taxon: Branchiopoda	Source: In-House Culture Age: <24
Sample ID: 09-7150-0428	Code: 39E7EB8C	Project: WET Compliance Testing
Sample Date: 12 Aug-19 07:10	Material: POTW Effluent	Source: City of Cadillac WWTP
Receipt Date: 13 Aug-19 11:00	CAS (PC):	Station: Outfall 001A
Sample Age: 30h (2 °C)	Client: City of Cadillac WWTP	

Data Transform	Alt Hyp	NOEL	LOEL	TOEL	TU	PMSD
Untransformed	C > T	50	76	61.64	2 1.6	30.48%

Dunnett Multiple Comparison Test

Per MDEGLE
KM

Control	vs	Conc-%	Test Stat	Critical	MSD	DF	P-Type	P-Value	Decision(α:1%)
Lab Water		13	-4.161	2.952	7.804	18	CDF	1.0000	Non-Significant Effect
		25	-1.816	2.952	7.804	18	CDF	0.9990	Non-Significant Effect
		50	2.307	2.952	7.804	18	CDF	0.0480	Non-Significant Effect
		76*	4.085	2.952	7.804	18	CDF	3.5E-04	Significant Effect
		100*	4.842	2.952	7.804	18	CDF	2.8E-05	Significant Effect

Test Acceptability Criteria

TAC Limits

Attribute	Test Stat	Lower	Upper	Overlap	Decision
Control Resp	25.6	15	>>	Yes	Passes Criteria
PMSD	0.3048	0.13	0.47	Yes	Passes Criteria

ANOVA Table

Source	Sum Squares	Mean Square	DF	F Stat	P-Value	Decision(α:5%)
Between	4295.28	859.057	5	24.58	<1.0E-37	Significant Effect
Error	1886.9	34.9426	54			
Total	6182.18		59			

Distributional Tests

Attribute	Test	Test Stat	Critical	P-Value	Decision(α:1%)
Variances	Bartlett Equality of Variance Test	1.764	15.09	0.8807	Equal Variances
Distribution	Shapiro-Wilk W Normality Test	0.9477	0.9459	0.0121	Normal Distribution

Reproduction Summary

Conc-%	Code	Count	Mean	95% LCL	95% UCL	Median	Min	Max	Std Err	CV%	%Effect
0	L	10	25.6	20.34	30.86	27	14	34	2.325	28.72%	0.00%
13		10	36.6	32.55	40.65	37	24	45	1.79	15.47%	-42.97%
25		10	30.4	26.61	34.19	30.5	22	40	1.675	17.42%	-18.75%
50		10	19.5	15.55	23.45	19.5	7	26	1.746	28.32%	23.83%
76		10	14.8	11.22	18.38	16	6	21	1.583	33.83%	42.19%
100		10	12.8	8.278	17.32	15.5	3	21	1.999	49.38%	50.00%

Reproduction Detail

Conc-%	Code	Rep 1	Rep 2	Rep 3	Rep 4	Rep 5	Rep 6	Rep 7	Rep 8	Rep 9	Rep 10
0	L	21	30	28	33	15	34	26	14	33	22
13		34	36	38	39	41	40	45	24	36	33
25		24	30	31	32	32	27	30	22	36	40
50		15	24	23	20	24	19	19	7	18	26
76		6	19	21	18	18	16	16	7	12	15
100		3	16	21	15	16	16	17	4	15	5

CETIS Analytical Report

Report Date: 19 Aug-19 16:11 (p 2 of 2)
Test Code/ID: 4236F9AA / 11-1089-9114

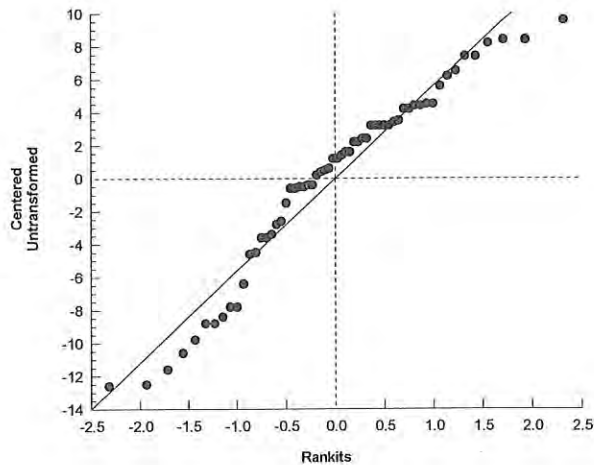
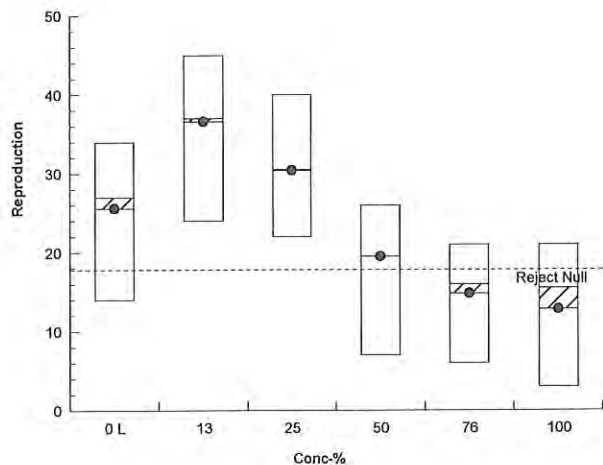
Ceriodaphnia 7-d Survival and Reproduction Test

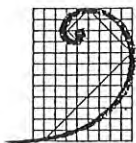
ERM

Analysis ID: 13-3335-8959 Endpoint: Reproduction
Analyzed: 19 Aug-19 16:09 Analysis: Parametric-Control vs Treatments

CETIS Version: CETISv1.9.4
Status Level: 1

Graphics





ERM®

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263

Phone: 616-399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

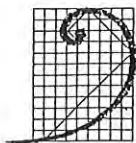
CLIENT NAME:	City of Cadillac				SAMPLER	TL				
ADDRESS:	1121 Platt Rd Cadillac, MI				PHONE NUMBER:	231-775-2368				
SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
Tert.	8-11-19	7:05AM		1 gal.	pH=7.26 NH ₃ = 4.3	0819-10	Temp. (°C) X On Ice	D.O. 10.1 mg/L	pH 7.3	Cond 995 umhos/cm
	8-12-19	7:10AM	comp.		pH= NH ₃ =		Temp. (°C) □ On Ice	D.O. mg/L	pH	Cond umhos/cm
					pH= NH ₃ =		Temp. (°C) □ On Ice	D.O. mg/L	pH	Cond umhos/cm
					pH= NH ₃ =		Temp. (°C) □ On Ice	D.O. mg/L	pH	Cond umhos/cm
					pH= NH ₃ =		Temp. (°C) □ On Ice	D.O. mg/L	pH	Cond umhos/cm
					pH= NH ₃ =		Temp. (°C) □ On Ice	D.O. mg/L	pH	Cond umhos/cm
ANALYSES REQUESTED [check item(s)]	Test Material: X Water/Wastewater Sediment Product		Test Type: X Acute X Chronic Other		Test Species: X Ceriodaphnia dubia Daphnia magna Daphnia pulex Fathead minnow (Pimephales promelas) Rainbow Trout (Oncorhynchus mykiss) Sheepshead minnow (Cyprinodon variegatus) Silverside minnow (Menidia beryllina) Other (write in comments section)					
COMMENT SECTION:										

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
T. Laczko	8-12-19	11:00 AM	K. Williams / ERM	8-19	1100

* See Instructions for Sample Collection on Back of Sheet

February 2018



ERM.

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263

Phone: 616-399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

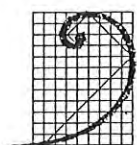
CLIENT NAME:	City of Cadillac				SAMPLER	72					
ADDRESS:	1121 Plett Rd Cadillac				PHONE NUMBER:	231-775-2841					
SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)				
Tert.	8-13-19	7:05 AM	Comp	1 gal	pH=7.36 NH ₃ =0.024 s.u. mg/L	081519-3	Temp. (°C) ☒ On Ice	D.O. mg/L 10.7	pH 7.6	Cond 974 umhos/cm	
	8-14-19	7:05 AM			pH= s.u. mg/L			Temp. (°C) ☐ On Ice	D.O. mg/L	pH	Cond umhos/cm
					pH= s.u. mg/L			Temp. (°C) ☐ On Ice	D.O. mg/L	pH	Cond umhos/cm
					pH= s.u. mg/L			Temp. (°C) ☐ On Ice	D.O. mg/L	pH	Cond umhos/cm
					pH= s.u. mg/L			Temp. (°C) ☐ On Ice	D.O. mg/L	pH	Cond umhos/cm
					pH= s.u. mg/L			Temp. (°C) ☐ On Ice	D.O. mg/L	pH	Cond umhos/cm
ANALYSES REQUESTED [check item(s)]	Test Material: ☒ Water/Wastewater ☐ Sediment ☐ Product		Test Type: ☒ Acute ☒ Chronic ☐ Other		Test Species: ☒ Ceriodaphnia dubia ☐ Daphnia magna ☐ Daphnia pulex ☐ Fathead minnow (Pimephales promelas) ☐ Rainbow Trout (Oncorhynchus mykiss) ☐ Sheephead minnow (Cyprinodon variegatus) ☐ Silverside minnow (Menidia beryllina) ☐ Other (write in comments section)						
COMMENT SECTION:											

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
T. Saizko	8-14-19	8:15 AM	Paul Wiggins / ERM	8-15-19	10:30

* See Instructions for Sample Collection on Back of Sheet

February 2018



ERM®

ENVIRONMENTAL RESOURCES MANAGEMENT

3352 128th Avenue Holland, Michigan 49424-9263

Phone: 616-399-3500 FAX: 616-399-3777

AQUATIC TOXICITY LAB CHAIN OF CUSTODY FORM *

CLIENT NAME:	City of Cadillac				SAMPLER	TL				
ADDRESS:	1121 Plott Rd Cadillac, MI				PHONE NUMBER:	231-775-2841				
SAMPLE DESCRIPTION (i.e. Outfall 001)	DATE (Begin End)	TIME (Begin End)	GRAB OR COMP	NUMBER AND SIZE OF CONTAINERS	FIELD PARAMETERS	SAMPLE ID NUMBER (Filled in by ERM)	INITIAL WATER QUALITY PARAMETERS UPON RECEIPT BY LABORATORY (filled in by ERM)			
Tert.	8-15-19	7:05 AM	Comp	1 gal	pH= 7.11 s.u. NH ₃ = .0716 mg/L	08/17/19-10	Temp. (°C) 10.2 On Ice	pH 7.1 s.u.	Cond 1017 umhos/cm	
	8-16-19	7:05 AM			pH= s.u. NH ₃ = mg/L			Temp. (°C) On Ice	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L			Temp. (°C) On Ice	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L			Temp. (°C) On Ice	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L			Temp. (°C) On Ice	pH s.u.	Cond umhos/cm
					pH= s.u. NH ₃ = mg/L			Temp. (°C) On Ice	pH s.u.	Cond umhos/cm
ANALYSES REQUESTED [check item(s)]	Test Material: ☒ Water/Wastewater ☐ Sediment ☐ Product		Test Type: ☒ Acute ☒ Chronic ☐ Other		Test Species: ☒ Ceriodaphnia dubia ☐ Daphnia magna ☐ Daphnia pulex ☐ Fathead minnow (Pimephales promelas) ☐ Rainbow Trout (Oncorhynchus mykiss) ☐ Sheepshead minnow (Cyprinodon variegatus) ☐ Silverside minnow (Menidia beryllina) ☐ Other (write in comments section)					
COMMENT SECTION:										

SAMPLE TRANSFERS

RELINQUISHED BY: Signature / Organization	DATE	TIME	ACCEPTED BY: Signature / Organization	DATE	TIME
T. Sazjko	8-16-19	9 AM	John P. Ren	08/17/19	1:00

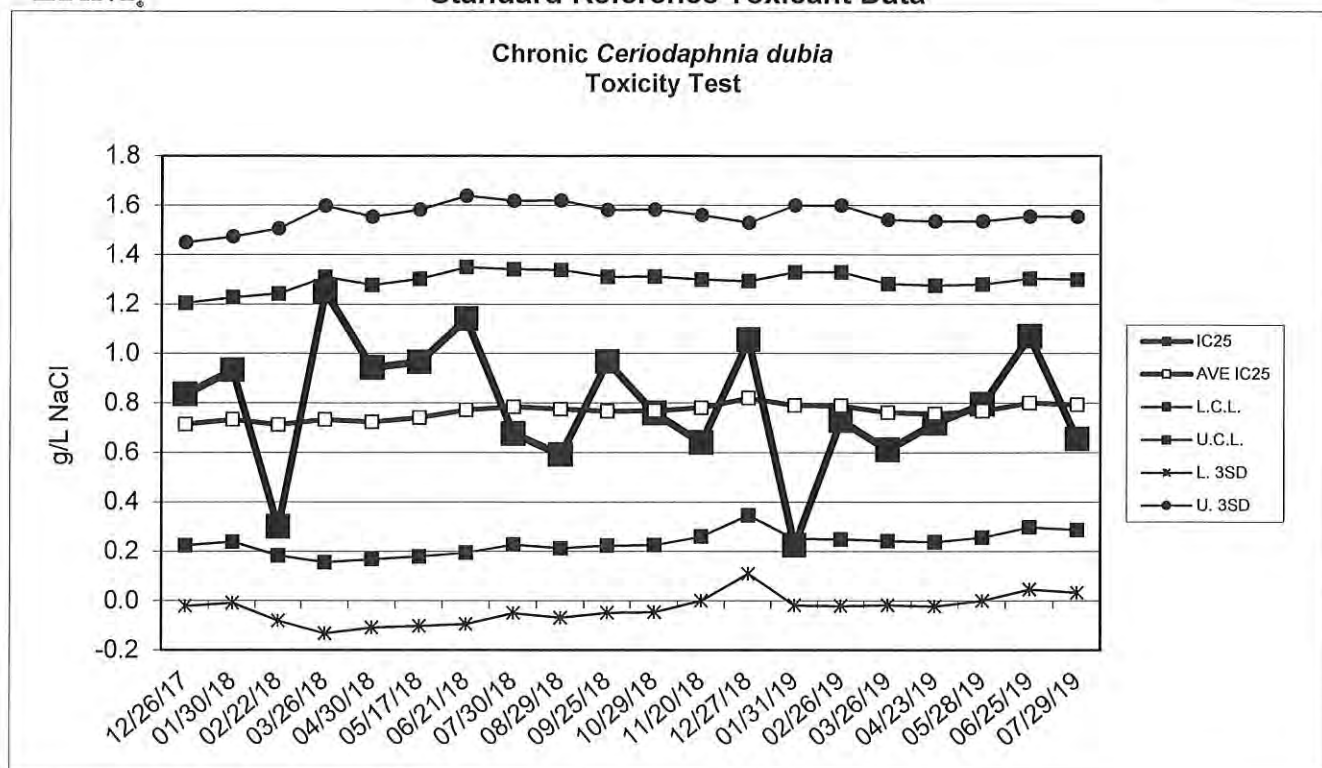
* See Instructions for Sample Collection on Back of Sheet

February 2018



Environmental Resources Management

Standard Reference Toxicant Data



Chronic *Ceriodaphnia dubia* Toxicity Test Data

Date	IC25 (g/L NaCl)	AVE IC25 (g/L NaCl)	CONTROL LIMIT		Survival (%)	CONTROL Reproduction (ave. young)	CV (%)
			Lower	Upper			
12/26/17	0.83	0.7	0.22	1.20	100	21.9	11.9
01/30/18	0.93	0.7	0.24	1.23	100	25.5	36.3
02/22/18	0.30	0.7	0.18	1.24	100	17.8	35.0
03/26/18	1.25	0.7	0.16	1.31	90	32.5	38.5
04/30/18	0.94	0.7	0.17	1.28	100	32.0	25.5
05/17/18	0.97	0.7	0.18	1.30	100	30.0	38.6
06/21/18	1.14	0.8	0.19	1.35	80	35.2	8.2
07/30/18	0.68	0.8	0.23	1.34	100	25.5	16.3
08/29/18	0.59	0.8	0.21	1.34	100	30.1	26.2
09/25/18	0.97	0.8	0.22	1.31	100	27.6	26.7
10/29/18	0.76	0.8	0.22	1.31	100	32.7	24.8
11/20/18	0.64	0.8	0.26	1.30	100	34.8	15.2
12/27/18	1.06	0.8	0.35	1.29	100	26.8	43.7
01/31/19	0.23	0.8	0.25	1.33	100	34.7	14.9
02/26/19	0.73	0.8	0.25	1.33	100	27.9	9.3
03/26/19	0.61	0.8	0.24	1.28	100	40.2	9.9
04/23/19	0.72	0.8	0.24	1.28	100	36.1	25.4
05/28/19	0.79	0.8	0.26	1.28	100	37.6	3.1
06/25/19	1.07	0.8	0.30	1.30	100	29.4	26.7
07/29/19	0.65	0.8	0.29	1.30	100	33.7	14.6

Appendix B
MDEGLE Reporting Form



MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

CERIODAPHNIA DUBIA CHRONIC TOXICITY TEST REPORT

By authority of PA 451 of 1994, as amended.

INSTRUCTIONS: Use this form to report chronic toxicity test results. Use separate forms for more than 1 test.

1. NAME OF FACILITY (on NPDES permit) City of Cadillac WWTP		2. NPDES PERMIT # M I 0 0 2 0 2 5 7							
1. RECEIVING WATER (as designated in permit) Clam River		2. OUTFALL 001A		5. RECEIVING WATER CONCENTRATION (if known)					
6. TEST LAB (Name and Address) Environmental Resources Management 3352 128 th Avenue Holland, MI 49424-9263									
7. TEST START DATE 08/13/19	8. TEST END DATE 08/19/19	9. AGE RANGE OF ORGANISMS AT TEST START 15 – 20.5 Hours Old				10. REPORT DATE 09/04/19			
11. NAME OF PERSON CONDUCTING TEST Lab Contact: Mark Baker		12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT Bruce Rabe (616) 738 – 7308							
13. SAMPLE COLLECTION DATES Sample 1: 08/12/19 Sample 2: 08/14/19 Sample 3: 08/16/19		14. DATE RECEIVED Sample 1: 08/13/19 Sample 2: 08/15/19 Sample 3: 08/17/19				15. ARRIVAL TEMP (C°) Sample 1: 2 Sample 2: 2 Sample 3: 1			
16. DATE OF FIRST USE Sample 1: 08/13/19 Sample 2: 08/15/19 Sample 3: 08/17/19		17. TOTAL RESIDUAL CHLORINE (mg/l) Sample 1: <0.01 Sample 2: <0.01 Sample 3: <0.01				18. AMMONIA (mg/l as N) Sample 1: <0.01 Sample 2: <0.01 Sample 3: <0.01			
19. WAS SAMPLE DECHLORINATED? Sample 1: ☐ YES ☒ NO IF YES ➡ Sample 2: ☐ YES ☒ NO IF YES ➡ Sample 3: ☐ YES ☒ NO IF YES ➡		20. DESCRIBE DECHLORINATION (if any)							
21. EFFLUENT SAMPLES WERE COLLECTED (check one) ☐ BEFORE CHLORINATION ☐ AFTER CHLORINATION ☐ AFTER CHLORINATION, BEFORE DECHLORINATION ☐ AFTER DECHLORINATION ☒ FACILITY DOES NOT CHLORINATE									
22. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded holding time.) N/A									
23. EFFLUENT FILTERED? ☐ YES ☒ NO IF YES ➡					24. STATE MESH SIZE OF FILTER				
25. EFFLUENT SAMPLE TYPE (check one type for each sample) Sample 1: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE Sample 2: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE Sample 3: ☒ 24-HR COMPOSITE ☐ GRAB/COMPOSITE (give # of grabs) ☐ GRAB SAMPLE						26. IDENTIFY THE DILUENT (O ₁) CONTROL <u>RMHW</u> IDENTIFY THE SECONDARY (O ₂) CONTROL (if used) _____			
27. SUMMARY OF DATA AND RESULTS - SURVIVAL AND REPRODUCTION									
CONCENTRATION OF EFFLUENT (%)	O ₁	O ₂	13%	25%	50%	76%	100%		
48-HOUR SURVIVAL (%)	100	N/A	100	100	100	100	100		
6-DAY MEAN REPRODUCTION/FEMALE	25.6	N/A	36.6	30.4	19.5	14.8	12.8		
6-DAY MEAN SURVIVAL (%)	100	N/A	100	100	100	100	100		
28. 48-HOUR LC ₅₀ (%) >100		29. TU _a (acute toxic units) 0							
30. 6-DAY CHRONIC VALUE (%) 61.6	31. NOEC 50%		32. LOEC 76%		33. TU _c (chronic toxic units) 1.6				

EQP 5819 (02/10)